PROJECT FLAMESCION

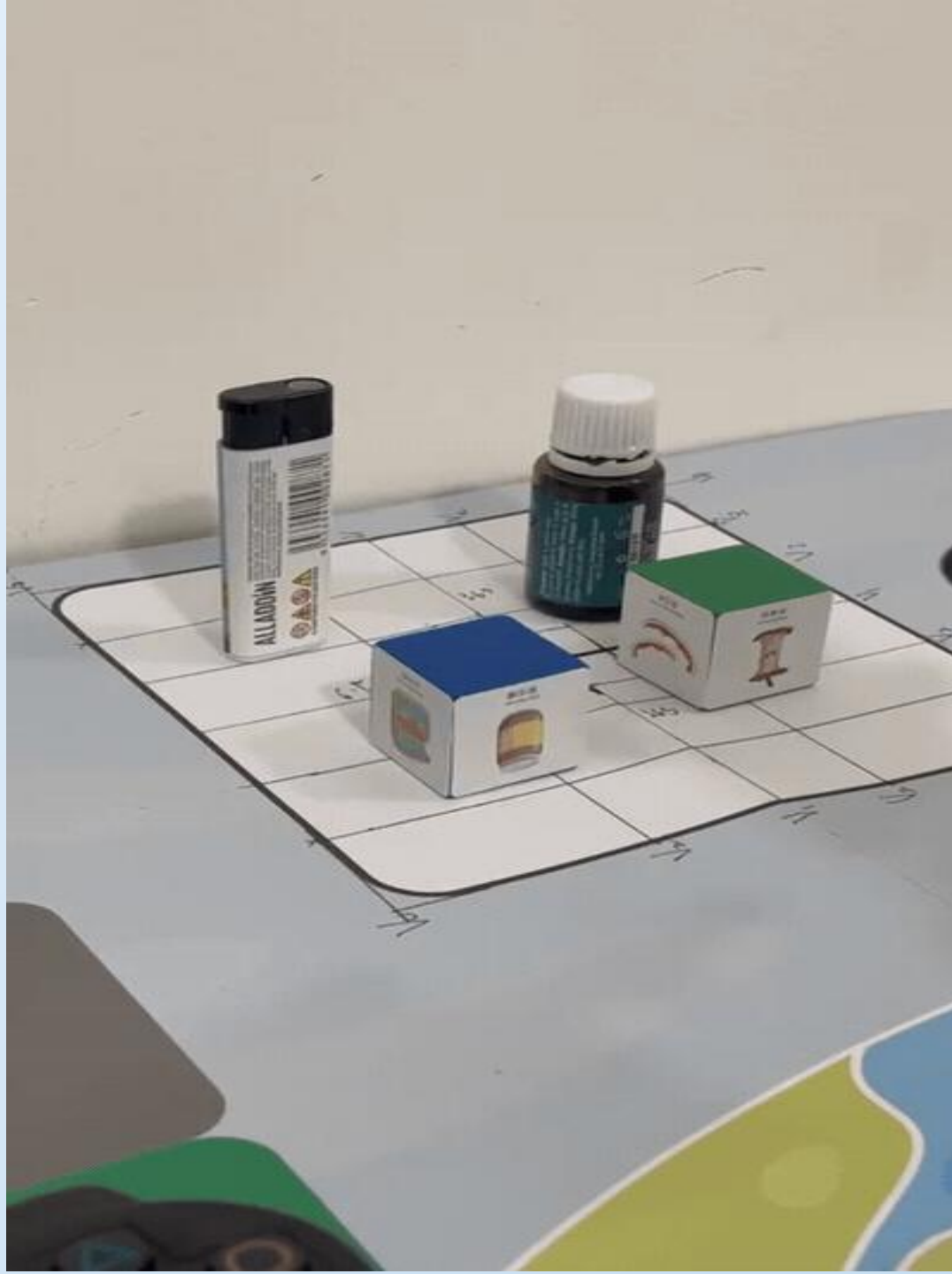
P300-TRANSFORMER BRAIN-COMPUTER INTERFACE FOR VISION-GUIDED ROBOTIC OBJECT RETRIEVAL

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APICTA 2026 | Tertiary Student Category

APICTA 2026



The Problem: Intention Without Movement

Aware of the room, unable to reach for anything in it



Intention ✓ → Physical action ✗

A person may remain aware of the environment while the normal muscular pathway to act is severely restricted.

- ALS
- Locked-In Syndrome
- Severe motor impairment

Benefit: Everyday Independence

Essential objects retrieved by attention alone

We know what we want

But we cannot reach it

- **Medicine** Taken on time
- **Water** Drink without waiting
- **Phone** Call for help
- **Personal item** Glasses, remote, keys



Benefit: Who Gains

The person, the family carer, and the clinic or lab

THE PERSON

Acts on their own choice

- Picks the real object they want
- No surgery, speech or hand movement
- Dignity and control in daily life

THE FAMILY CARER

Fewer small requests

- The arm handles routine fetch tasks
- Time back for care that needs a person
- The user can act while the carer rests

CLINICS AND LABS

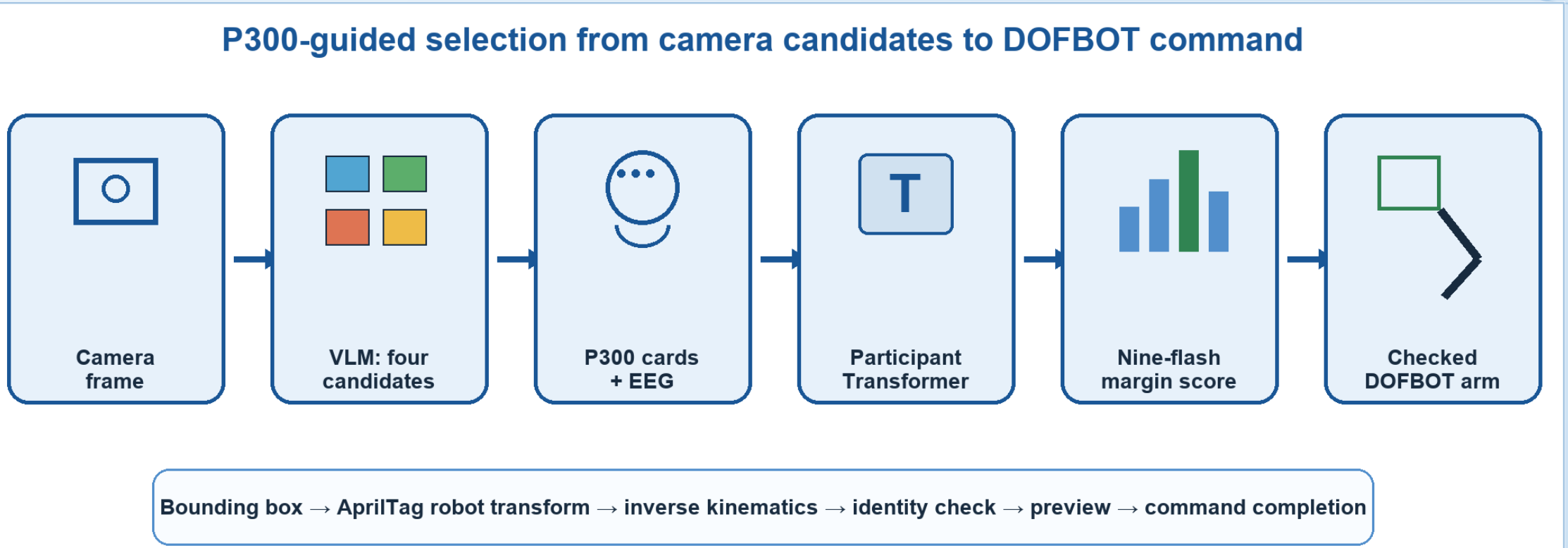
Deployable station

- Non-invasive, reusable saline cap
- New objects without retraining
- Hands-free picks on sterile benches

Why it matters: ALS caregiving brings long daily hours, fatigue and lost household income (Pagnini et al. 2010; Schischlevskij et al. 2021).

How Flamescion Works: Eye, Brain, Arm

One connected path from a visible object to a robot pick



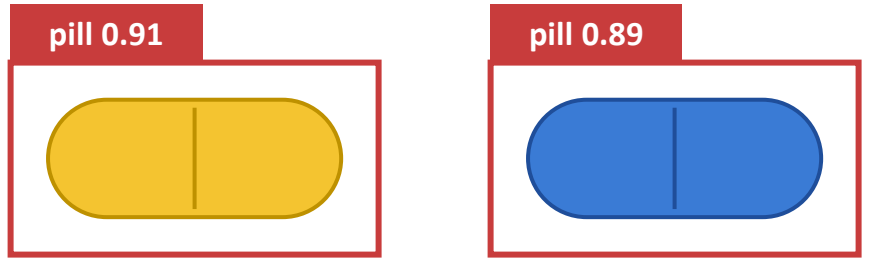
Scene → intention → EEG → Transformer → object decision → verified robot action

Uniqueness: Context Matters, Not Just Class

The vision-language model names each object, not just its class

TRADITIONAL DETECTOR, e.g. YOLO

A fixed class list



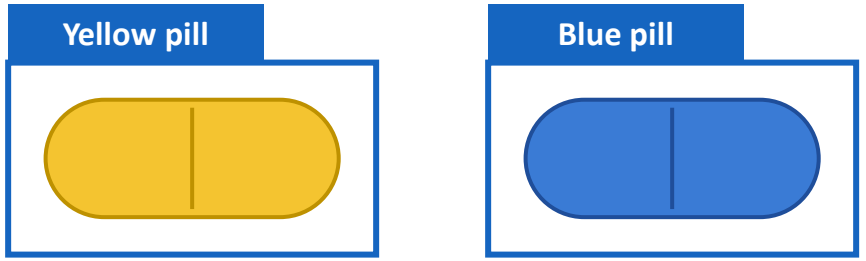
Both cards read "pill"

Same class, same label. The user cannot tell which pill is which.

Knows only the classes it was trained on

FLAMESCION VISION-LANGUAGE MODEL

Understands context



Cards read "Yellow pill" and "Blue pill"

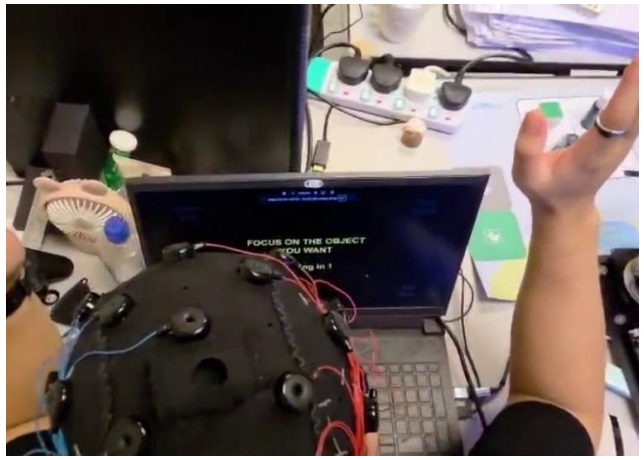
Name, colour and description for each object, so the user picks the exact one.

Fine-tuned on Malaysian and medical objects: 83% to 93%

Context matters: with medicine, picking the right pill is the whole point.

Uniqueness: The Brain Controls a Robotic Arm

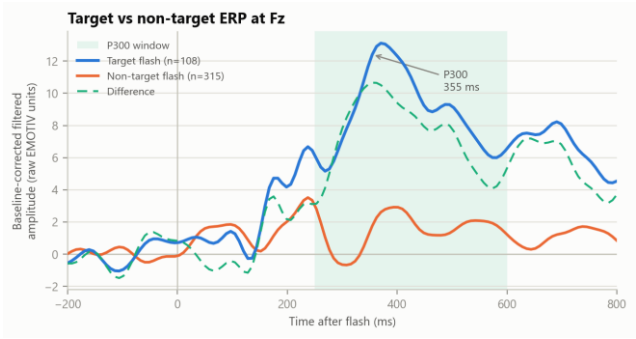
Thought in, physical action out: no hands, no voice, no surgery



1 ATTEND

The user focuses on the object

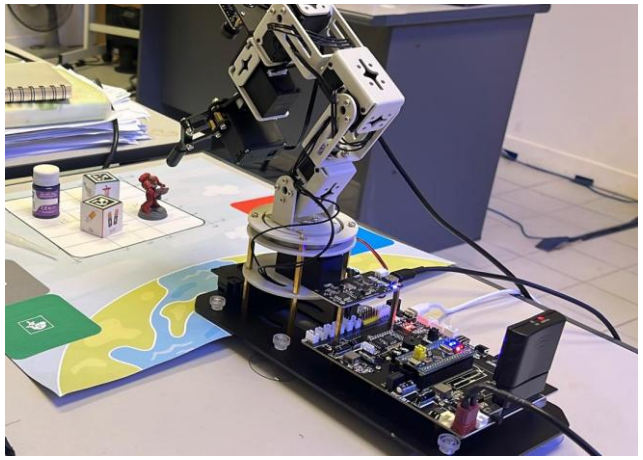
32-channel saline EEG cap reads the brain



2 DECODE

The brain's P300 picks it

Transformer finds the response 0.3 s after the flash



3 ACT

The robotic arm retrieves it

DOFBOT picks up the chosen object

A P300 speller ends in text on a screen. Flamescion ends with a robotic arm picking up the object.

Uniqueness: Nine Flashes, One Choice

A 1.74 M-parameter Transformer pools repeated brain evidence



1 FLASH

One flash gives one margin.

target logit – non-target logit

2 REPEAT

Nine margins are gathered for each object.

one margin from each repetition

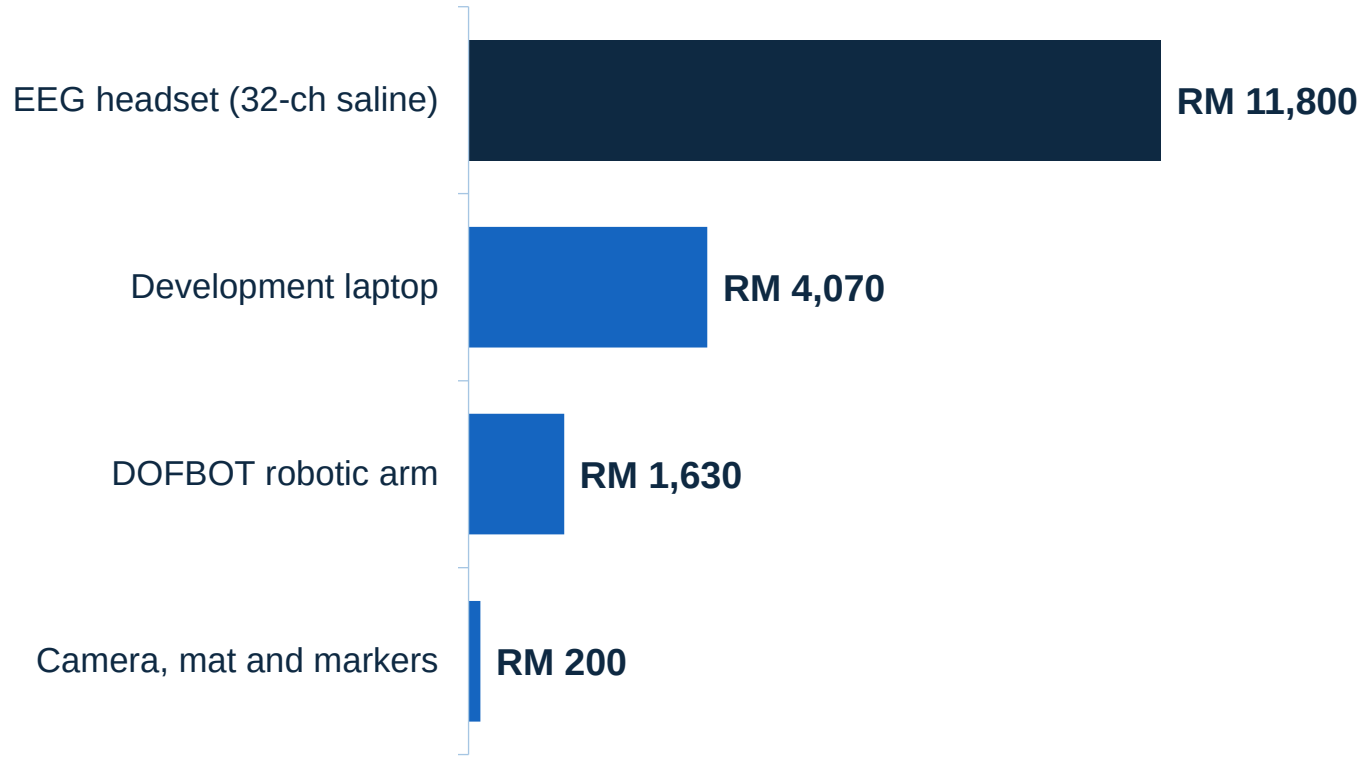
3 SELECT

Mean margins are compared across the four objects.

the highest mean wins

Value: Affordable to Build and Run

One-time hardware dominates; each live selection costs under RM 0.05



~RM 17,700

complete station, one-time

< RM 0.05

per live selection

~RM 41–81

per 8.5 h model-training run

Indicative retail pricing at 1 USD = 4.07 MYR (28 Sep 2026). The EEG headset is two-thirds of the total.

Value: Better Days for User and Carer

What changes when small daily tasks no longer need a helper

A DAY WITH FLAMESCION

8:00
Morning medicine
taken on time

12:30
A drink
without calling anyone

15:00
The phone
to call family

21:00
Glasses or remote
before sleep

FOR THE USER

Independence that protects dignity

- Acts on their own choice, keeping dignity and self-esteem (Chochinov et al. 2002)
- Less guilt about the strain on family (Aoun et al. 2011)

FOR THE FAMILY CARER

Room to rest, work and live

- Fewer small requests in long care days (Pagnini et al. 2010)
- More time for work and social life, often lost to care (Oh et al. 2021; Schischlevskij et al. 2021)

Value: What Makes Flamescion Stand Out

The one route here that meets all four needs of a user who cannot move

	Eye-gaze board	Implanted BCI	P300 speller	Flamescion
Non-invasive, no surgery	✓	✗	✓	✓
No speech or gaze-dwell control	✗	✓	✓	✓
Choices drawn from the live scene	✗	✗	✗	✓
Ends in a physical object retrieval	✗	✓	✗	✓

PRACTICAL VALUE

Laptop-class hardware
Decoder runs on a 4 GB laptop GPU

Swap objects freely
Cards carry names, so no retraining

Every block logged
Each decision can be audited

Beyond the bedside
Hands-free work on sterile benches

Value: Ready to Scale

Each new user, object, room or device is a small, known step

01

New user

Five short EEG sessions (30–40 min) and one cloud training run of about 8.5 h, RM 41–81.

02

New objects

No retraining. The vision model names whatever is on the table.

03

New room

Re-anchor the arm with the calibration mat and one AprilTag marker.

04

New hardware

Eye, brain and arm are separate modules. The arm has already been swapped once, from an ESP32 prototype to the DOFBOT.

Built on a public base model of 24,091 EEG epochs, so each person only adds their own sessions.

Value: Sustainable and SDG-Aligned

Reusable hardware, light daily energy use, four UN goals served

SUSTAINABILITY

01

Reusable saline cap

Salt-water electrodes, cleaned and reused: no single-use gel waste

02

Train once, in the cloud

One 8.5 h run on a single GPU per campaign, then switched off

03

Run on a laptop

Daily decoding on a 4 GB laptop GPU, no always-on server

UN SUSTAINABLE DEVELOPMENT GOALS

SDG 3

Good Health and Well-being

Assistive action without a manual interface

SDG 9

Industry, Innovation and Infrastructure

Neural decoding, vision and robotics in one pipeline

SDG 10

Reduced Inequalities

Access for people who cannot use standard inputs

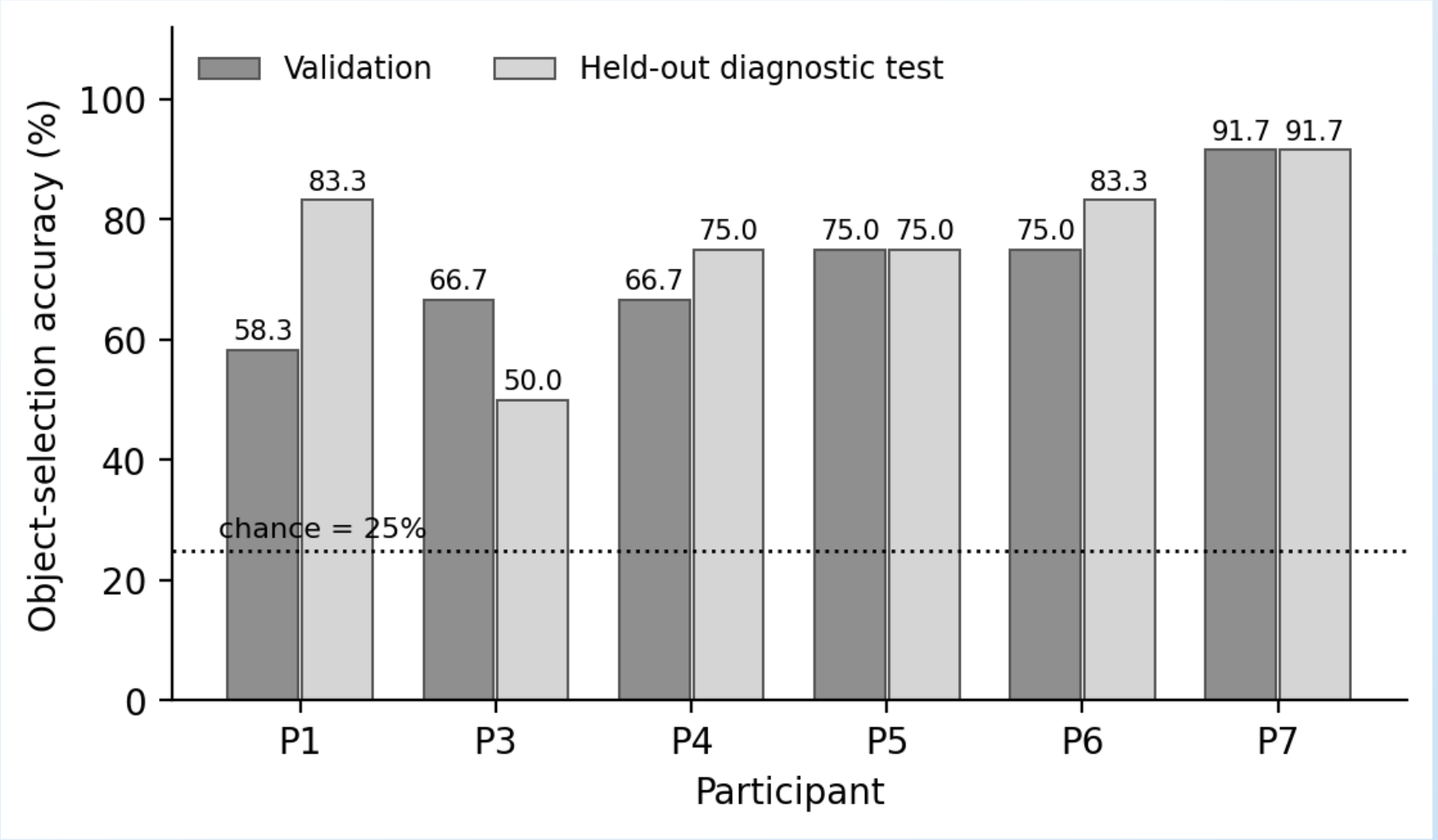
SDG 12

Responsible Consumption and Production

Reusable cap and low-power daily operation

Evidence: Accurate in Testing and Live

Saved per-participant checkpoints on held-out sessions; four-choice chance = 25%



Results

91.67%

best held-out (P7)

71.43%

live selection, 35 / 49

Robot: 48 / 68

Result: Over 70% Live, End to End

Measured across the whole loop: camera, brain and arm

LIVE BCI SELECTION

71.43%

35 of 49 labelled live decisions correct



ROBOT EXECUTION

70.59%

48 of 68 commands completed by the arm



Almost three times the 25% chance level of a four-object choice.

WHY IT DESERVES TO WIN

Why Flamescion Deserves to Win

01

Solves a real gap

Attention alone becomes a physical object retrieval.

02

Stands out

Live-scene choices, a P300 Transformer and a checked arm.

03

Proven in use

91.67% held-out selection; live robot retrieval at ~70%.

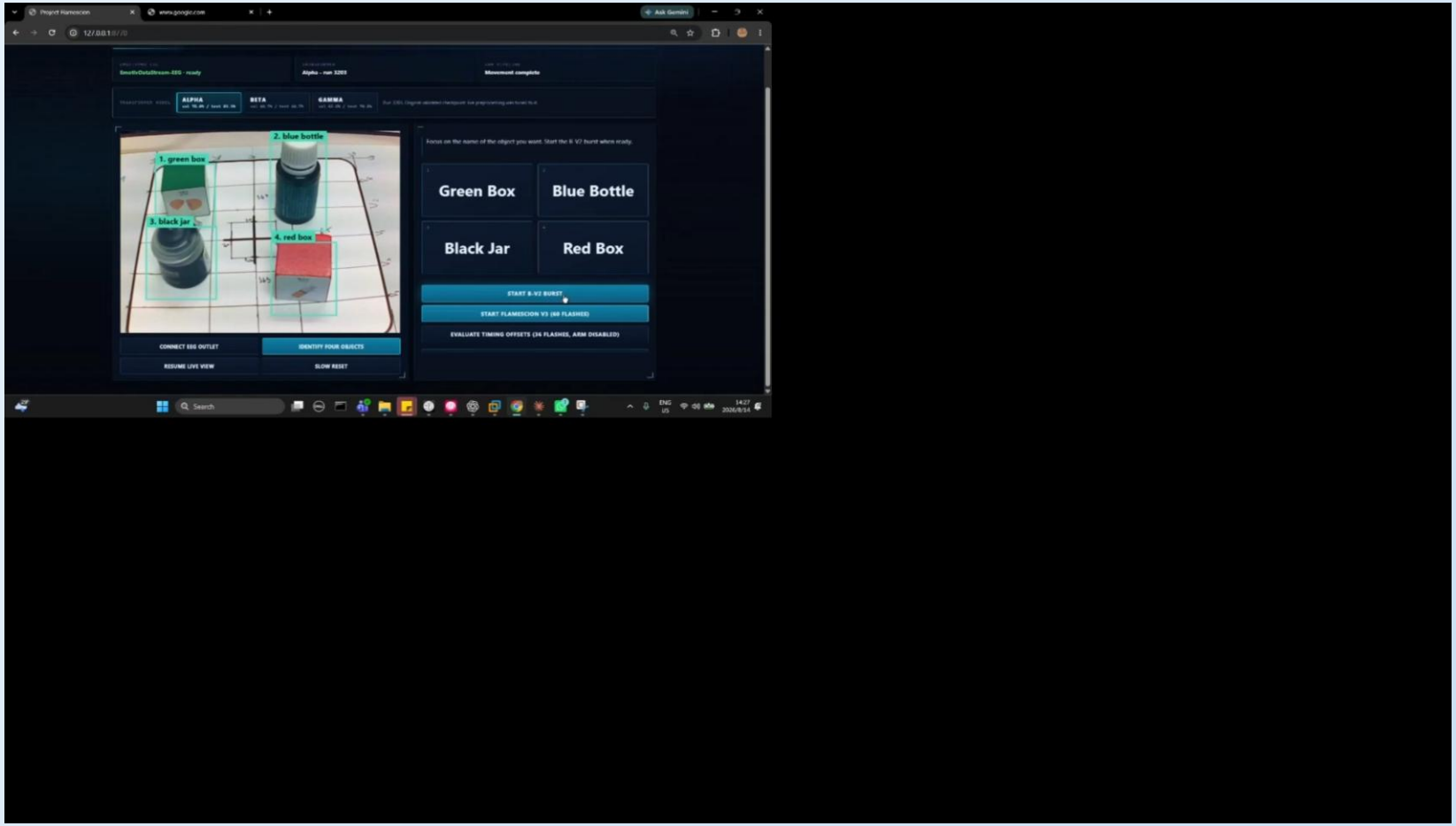


Product Demonstration

Recorded end-to-end run: select, verify, preview, execute

1. selected
2. verified
3. preview
4. execute

Robotic Arm in execution



Recorded live run, sped up to about 35 s